Assignment No-05

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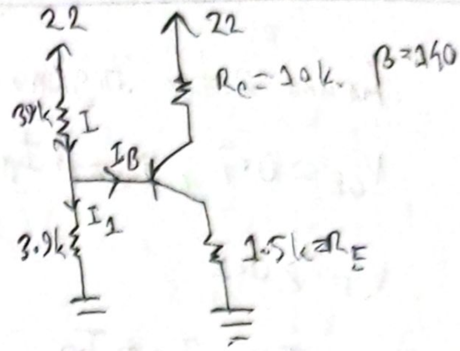
Course : CSE251.

Section : 22.

Ans to the Q No-1



Assume active region, $V_{BE} = 0.7$, $I_C = \beta I_B$, $I_C = \alpha I_E$

We have to verify $V_{CE} > 0.2$.

Using KVL, $22 = 39I + 0.7 + 1.5 I_E$

$\Rightarrow 22 = 39I + 0.7 + 1.5(141) I_B$

$\Rightarrow 39I + 211.5 I_B = 21.3$ --- (i)

$\beta I_B = \alpha I_E$

$\Rightarrow I_E = \frac{\beta}{\alpha} I_B$

$\Rightarrow I_E = (1+\beta) I_B$

Now Using KVL, $22 = 39I + 39I_1$

$\Rightarrow 22 = 39I + 3.9(I - I_B)$ [$I = I_B + I_1$ using KCL]

$\Rightarrow 22 = 39 + 3.9I - 3.9I_B$

$\Rightarrow 42.9I - 3.9I_B = 22$ --- (ii)

Solving equation (i) & (ii), $I = 0.51$ mA & $I_B = 6.04$ µA.

Now, $I_C = 140 \times 6.04 = 0.845$ mA

$I_E = 141 \times 6.04 = 0.852$ mA

Now, Using KVL

$22 = I_C R_C + V_{CE} + I_E R_E$

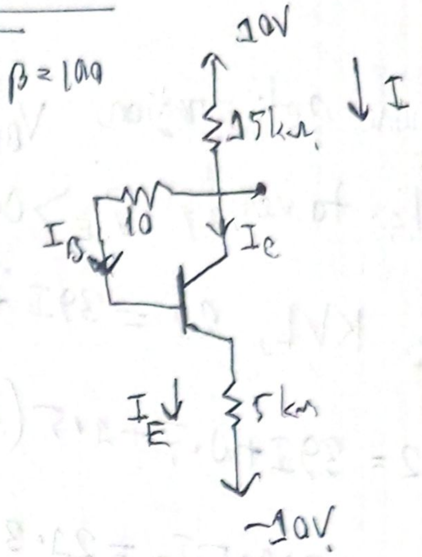
$\Rightarrow V_{CE} = 15 - (0.845 \times 10) - (0.852 \times 1.5) = 12.272$ V.

As $V_{CE} > 0.2$ [Assumption correct]

So, $V_{CE} = 12.272$ V and $I_C = 0.845$ mA

Ans

Ans to the Q No-2



Assume active region,

$V_{BE} = 0.7$, $I_c = \beta I_B$ and we have to verify

$V_{CE} > 0.2$.

Now, $I = I_B + I_C$

Using KVL, $10 = 15I + 10 I_B + V_{BE} + 5 I_E - 10$.

$\Rightarrow 10 = 15(I_B + I_C) + 10 I_B + 0.7 + 5(1+\beta) I_B - 10$

$\Rightarrow 19.3 = 15 I_B + 15\beta I_B + 10 I_B + 5 I_B + 5\beta I_B$

$\Rightarrow 19.3 = 15 I_B + 1500 I_B + 10 I_B + 5 I_B + 500 I_B$

$\Rightarrow 2030 I_B = 19.3$

$\therefore I_B = 9.507 \times 10^{-3}$ mA

Now,

$I_C = 100 \times 9.507 \times 10^{-3} = 0.95$ mA

$I_E = 101 \times 9.507 \times 10^{-3} = 0.96$ mA

So, $I = 9.507 \times 10^{-3} + 0.95 = 0.96$ mA.

Now,

$$\frac{10 - V_C}{15} = 0.96$$

$\therefore V_C = -4.4$ V.

Now, $\frac{V_E + 10}{5} = 0.96$

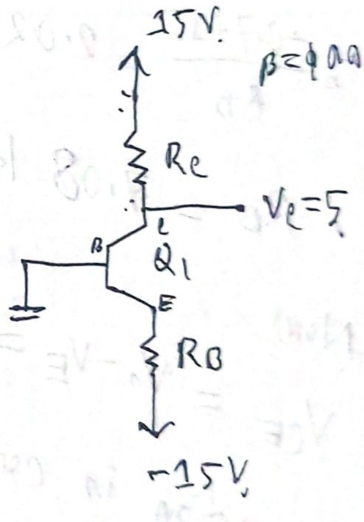
$\Rightarrow V_E = -5.2$ V.

So $V_{CE} = V_C - V_E = -4.4 - (-5.2) = -4.4 + 5.2 = 0.8$ V.

As $V_{CE} > 0.2$ so assumption correct.

So, $I_C = 0.95$ mA and $V_C = -4.4$ V. Ans

Ans to the Q No-3



Here, $I_C = 2$ mA assume it is in active region,

$V_{BE} = 0.7$ and we have to verify $V_{CE} > 0.2$.

Now $\frac{15 - 5}{R_C} = 2$ $\left[\frac{15 - V_C}{R_C} = I_C\right]$

$\Rightarrow R_C = 5\,k\Omega$.

Now, $I_C = \beta I_B$

$\therefore I_B = \frac{I_C}{\beta}$

$\therefore I_B = \frac{2}{100}$

$\therefore I_B = 0.02$ mA.

Now, $I_E = (1+\beta) I_B = 101 \times 0.02 = 2.02$ mA.

Here $V_B = 0$ V.

$V_{BE} = 0.7$.

$\Rightarrow V_B - V_E = 0.7$,

$\Rightarrow V_E = -0.7$.

Now

$$\frac{V_E - (-15)}{R_B} = I_E$$

$$\Rightarrow \frac{-0.7 + 15}{R_B} = 2.02$$

$\Rightarrow R_B = 7.08$ kΩ.

Now

$V_{CE} = V_C - V_E = 5 + 0.7 = 5.7$ which in greather than 0.2. So our assumption in correct.

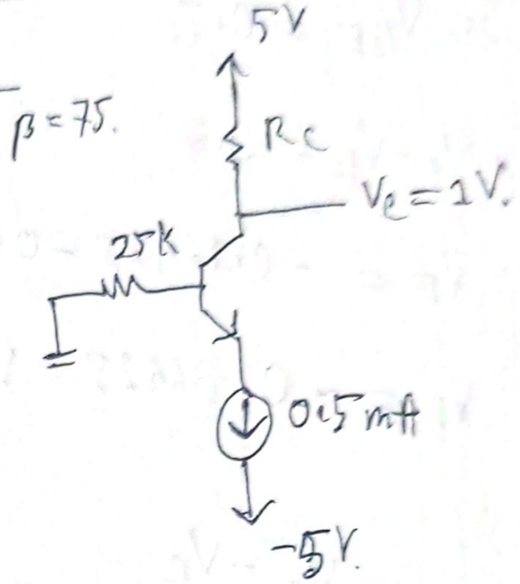
So,

$R_C = 5$ kΩ.

$R_B = 7.08$ kΩ.

Ans

Ans to the Q No-4



Assume active mode, so, $V_{BE}=0.7$. so we have to verify $V_{CE}>0.2$.

Here,

$$I_E = 0.5\,mA$$

$$I_c = \alpha I_E$$

$$I_c = \frac{\beta}{1+\beta} I_E$$

$$\therefore I_c = \frac{75}{76} \times 0.5$$

$$\therefore I_c = 0.493\ mA.$$

So

$$I_B = \frac{0.493}{75} = 6.573\times10^{-3}\,mA.$$

Now

$$\frac{5-V_c}{R_c} = I_c$$

$$\therefore R_c = \frac{5-1}{0.493}$$

$$\therefore R_c = 8.113\ k\Omega.$$

Now,

$$\frac{0-V_B}{25} = 6.57\times10^{-3}$$

$$\therefore V_B = -0.16425\ V.$$

So, $V_{BE} = 0.7$.

$\Rightarrow V_B - V_E = 0.7$

$\Rightarrow V_E = -0.16425 - 0.7$.

$\Rightarrow V_E = -0.86425 \text{ V}$.

So

$V_{CE} = V_C - V_E$

$= 1 - (-0.86425)$

$= 1 + 0.86425$

$= 1.86425 \text{ V}$.

So,

$V_{CE} > 0.2$ assumption is correct.

$R_C = 8.113 \text{ k}\Omega$.

(Ans)